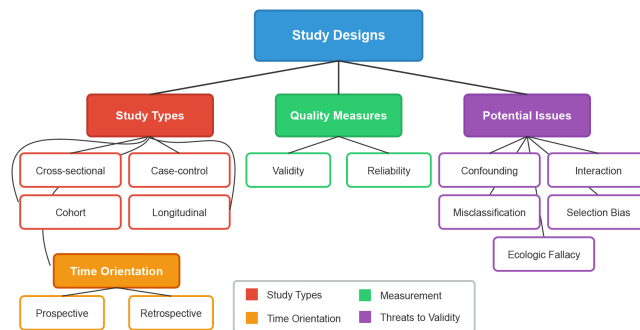


Designing Population Health Studies

April 6 and April 8 Eric Delmelle



Chapter Overview

- Research design fundamentals in epidemiology and public health
- Measurement validity and reliability
- Error and bias in population health research
- Study design approaches:
 - Cross-sectional
 - Case-control
 - Cohort
 - Experimental

- Qualitative methods and their integration with quantitative approaches
 - Ethical considerations in population health research
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1 A Matter of Measurement

Primary vs. Secondary Data

Primary data:

- Data collected specifically for the purpose of the study.

Secondary data:

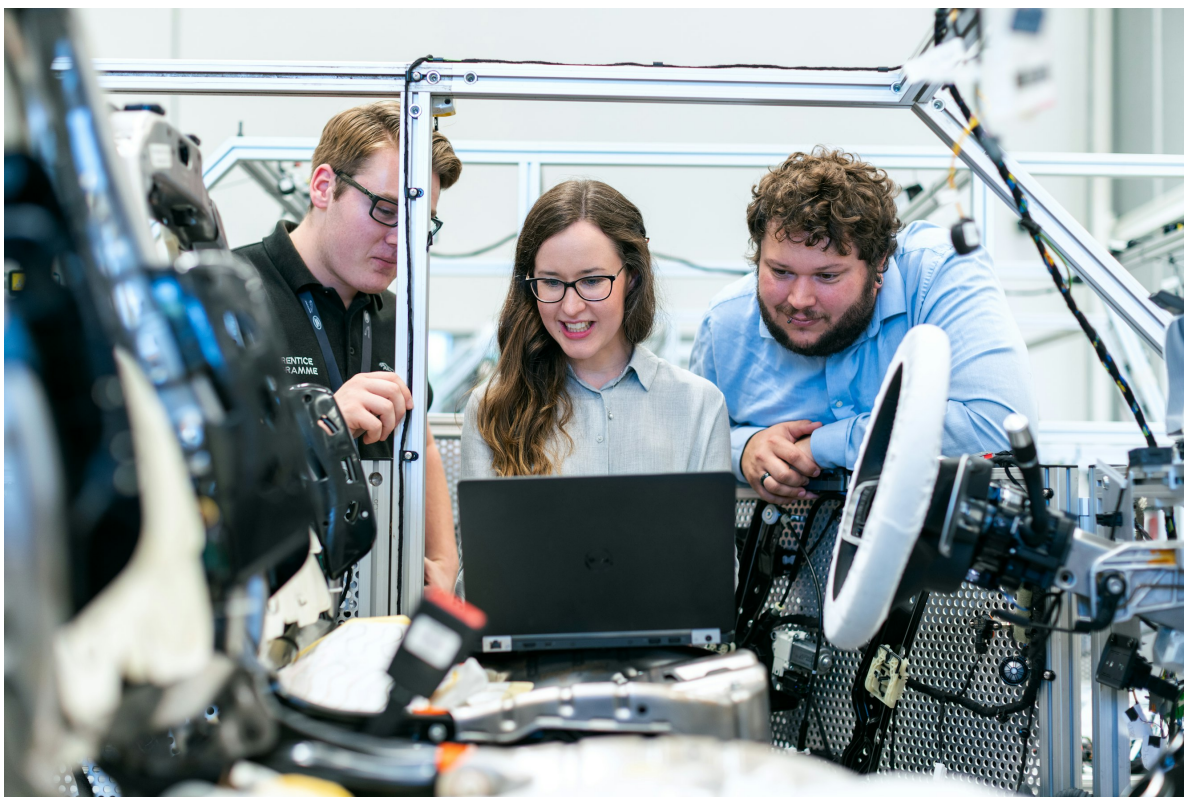
- Data collected for other purposes but reorganized and reanalyzed.

Examples:

- Health insurance claims data
 - Employment records
 - National health surveys
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2 Primary vs. Secondary Data

- **Primary data**
 - Collected specifically for a new study
 - Controlled by the researcher
 - Tailored to answer specific research questions
- **Secondary data**
 - Pre-existing data collected for another purpose
 - Often large-scale and readily available
 - May require cleaning or transformation



While primary data offers control and precision, secondary data can save time and resources.

3 Levels of Measurement

Level	Description	Example	Operations
Nominal	Categories with no ranking	Blood type, Sex	Equality/inequality
Ordinal	Ordered categories	Health self-rating: excellent, good, fair, poor	Greater than/less than
Interval	Equal units, no true zero	Temperature in °C	Addition/subtraction
Ratio	Equal units with true zero	Weight, Blood pressure	Multiplication/division

Understanding measurement levels is crucial for selecting appropriate statistical analyses. A

variable can always be reduced to a lower level of measurement (continuous to categorical), but not elevated (categorical to continuous).

4 Ecological Studies and Fallacy

- **Unit of analysis:** Group (e.g., city-level data)
- **Examples:**
 - Community fluoride levels and dental caries
 - Countries' smoking rates and lung cancer rates
- **Ecological fallacy:** Attributing group-level associations to individuals
- **Example:**
 - Classrooms with more women had higher average grades
 - But individual-level analysis showed men had higher grades in each classroom

	Classroom A	Classroom B	Classroom C
F 70	F 65	F 65	
F 70	F 70	F 70	
F 70	F 70	F 80	
F 75	F 75	F 80	
F 70	F 80	M 70	
F 80	F 85	M 75	
F 80	M 80	M 75	
F 80	M 80	M 80	
M 95	M 85	M 85	
M 100	M 90	M 90	
Class Mean			
F 74, M 98, FM: 79	F 74, M 84, FM: 78	F 74, M 79, FM: 77	

5 Variables and Levels of Measurement

- **Categorical variables:**
 - *Dichotomous* (e.g., male/female)
 - *Polytomous* (e.g., blood type)
 - *Nominal* (no implied order)
 - *Ordinal* (ranked, e.g., “good” > “fair”)

- **Continuous variables:**
 - *Interval scale* (e.g., temperature in Celsius)
 - *Ratio scale* (e.g., body weight, height)

Note: Continuous variables can be converted to categorical, but not vice versa.



6 Types of Research Design

Key concepts to distinguish studies:

- **Purpose:** Descriptive vs. analytical
- **Investigator control:** Observational vs. interventional
- **Directionality:** Forward vs. backward
- **Sample selection:** Based on exposure, disease, or neither
- **Timing:** Prospective vs. retrospective

Study Types: - Cross-sectional - Case-control - Retrospective cohort - Prospective cohort - Randomized controlled trial (RCT)

7 Basic Terminology: Exposure and Disease

- **E** = “Exposure”
 - Risk factor (e.g., smoking, occupational hazard)
 - Intervention (e.g., drug, prevention program)
- **D** = “Disease” or outcome
 - Disease, injury, death
 - Any health-related outcome
- **E** and **D** = Absence of exposure/disease
- **E** and **D** = Presence of exposure/disease



This basic E and D framework helps illustrate the logic behind different epidemiological study designs.

8 Study Designs Summary

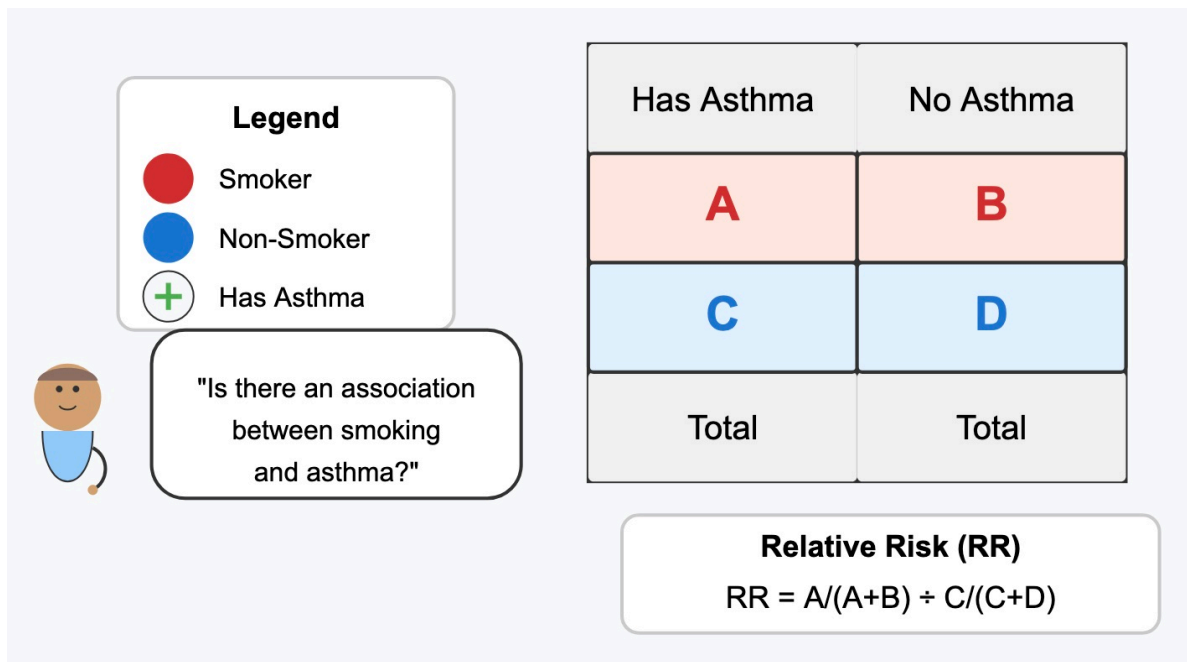
Study Type	Purpose	Control	Directionality	Sample Selection	Timing
Cross-sectional	Descriptive/Analytical	Observational	Concurrent	Representative sample	Retrospective
Case-control	Analytical	Observational	Backward	Based on disease	Retrospective
Retrospective cohort	Analytical	Observational	Forward	Based on exposure	Retrospective
Prospective cohort	Analytical	Observational	Forward	Based on exposure	Prospective
Randomized control trial	Analytical	Interventional	Forward	Based on exposure	Prospective

9 Cross-Sectional Studies

- Also called **prevalence studies**
- Exposure and outcome assessed **simultaneously**
- Can be descriptive or analytical
- Provides a “snapshot” of a population
- Relatively quick and inexpensive

Limitations:

- Cannot establish temporal sequence
- Only includes survivors of disease
- Not suitable for rare diseases



A snapshot in time: both exposure and outcome measured simultaneously

10 In-class exercise

11 Case-Control Analysis

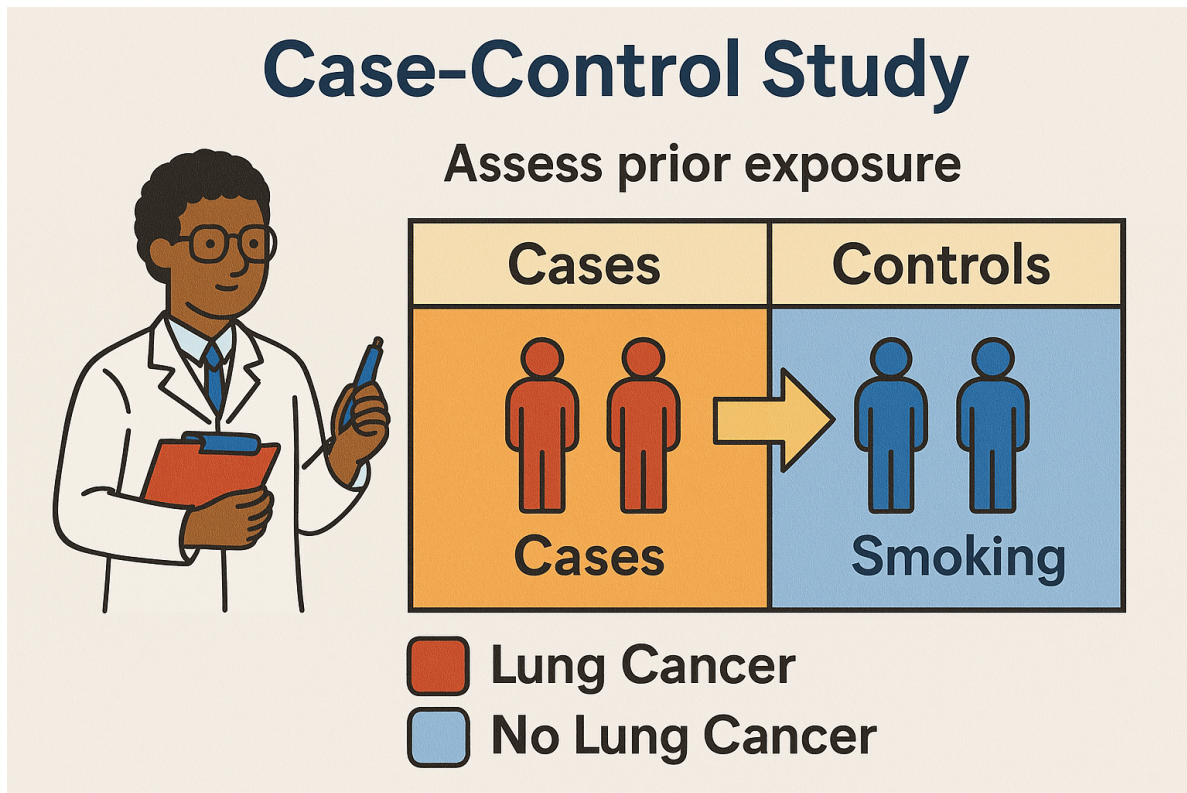
- Cannot directly compute relative risk
- Use **odds ratio (OR)** as an estimate:

$$OR = \frac{ad}{bc}$$

- In a 2×2 table:

	D_1	D_0
E_1	a	b
E_0	c	d

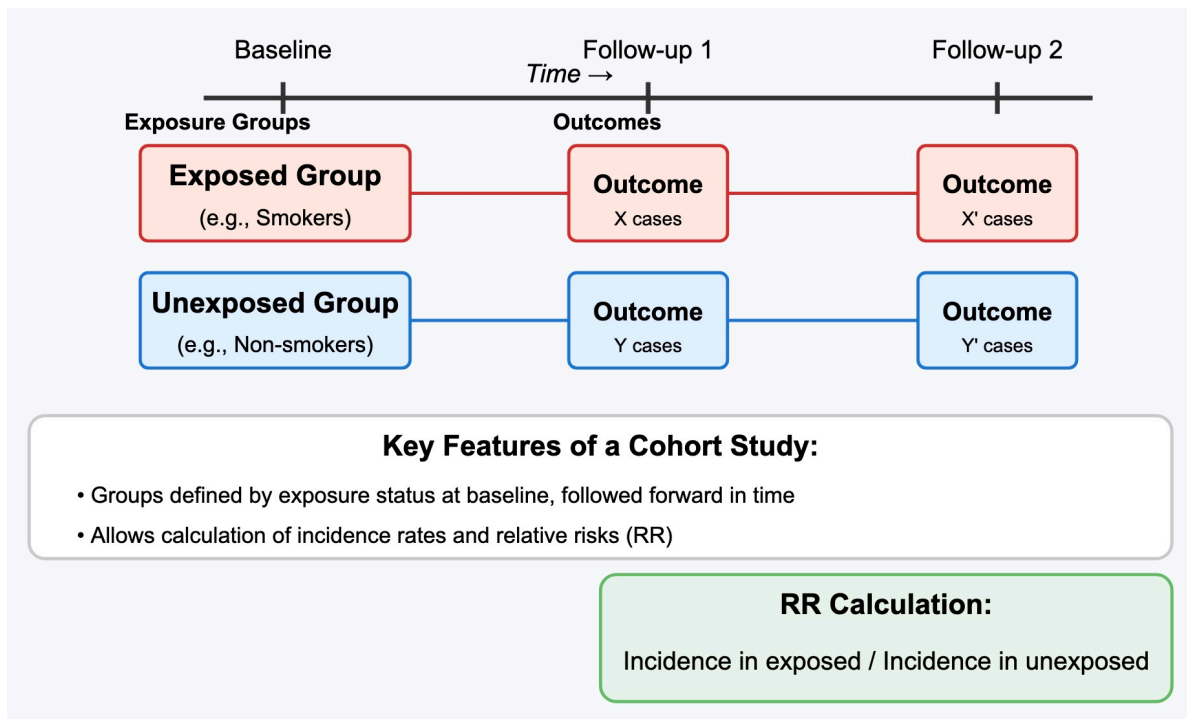
- When disease is rare, OR = RR



What is happening here?

12 Cohort Studies

- Start with **exposure status** (E and E)
- Follow forward to observe outcome
- Two types:
 - **Prospective:** Start now, follow into future
 - **Retrospective:** Look back at historical exposure
- Can study multiple outcomes
- Directly computes incidence and relative risk



Following groups forward from exposure to outcome

13 Cohort Analysis

- **Relative Risk (RR)** quantifies association between exposure and outcome:

$$RR = \frac{a/(a+b)}{c/(c+d)}$$

- In a 2×2 table:

	D_1	D_0
E_1	a	b
E_0	c	d

- Allows for:
 - Direct incidence calculation
 - Assessment of multiple outcomes

- Use with rare exposures

Challenges:

- Time-consuming and costly
 - Potential for loss to follow-up
 - Not ideal for rare diseases
 - Diagnostic changes over time may affect results
-

14 Randomized Controlled Trials (RCTs)

- **Gold standard** for assessing causal relationships
- Participants randomly allocated to intervention (E) or control (E)
- Minimizes confounding and bias through:
 - Randomization
 - Blinding (single, double)
- Can be conducted at individual or group level

Phases of clinical trials:

- **Phase I:** Safety and dosage (small sample)
- **Phase II:** Effectiveness and side effects
- **Phase III:** Confirm effectiveness, monitor adverse reactions (RCT)
- **Phase IV:** Post-marketing surveillance

Randomization helps balance known and unknown confounders.
